

PROJECT DOCUMENTATION REPORT: APPLYNOW

Project Title: ApplyNow

Role: Frontend Developer

Type: Tertiary Admissions & Document Verification Web Platform

1. Project Overview & Introduction

ApplyNow is an automated admission screening and applicant tracking platform built to modernize tertiary institution admissions in Nigeria (JAMB UTME, WAEC, NECO). The platform simplifies onboarding for prospective students, automates document scanning, and gives admissions officers an intelligent evaluation dashboard.

My Role: As the Frontend Developer, I architected the user interface, managed multi-step application state, integrated Google Gemini Multimodal Vision API for client-side document OCR, implemented deterministic admission calculation logic, and ensured zero-error code quality and responsive design.

2. Problem Statement (Pain Points Solved)

- **Manual Data Entry Errors:** Applicants frequently mistype registration numbers, exam years, and composite scores, creating administrative bottlenecks.
 - **Lack of Transparency:** Students rarely understand how their O'Level grades and UTME scores are combined, leading to confusion and high support inquiries.
 - **Disorganized & Slow Record Processing:** Admissions boards spend weeks manually cross-referencing physical result sheets and computing cutoff aggregates across departments.
 - **Applicant Anxiety:** Lack of real-time application tracking leaves candidates uncertain about their admission standing.
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3. Tech Architecture & Toolset

- **Frontend Framework:** React 19 (Component-driven UI, fast rendering)
- **Build Tool:** Vite 8 (Instant Hot Module Replacement, optimized bundling)
- **Styling:** Tailwind CSS v4 (Modern, utility-first responsive layout)

- **Form & Validation:** Formik with Yup (Strict schema validation for registration numbers and score boundaries)
 - **AI & OCR Integration:** Google Gemini Multimodal Vision API (gemini-3.6-flash, gemini-3.5-flash) for instant document credential scanning
 - **Routing & State:** React Router DOM v7 and React Context API (persistent multi-step onboarding data)
 - **Icons & UI Effects:** FontAwesome SVG Core and Framer Motion animations
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4. Core Features & Implementation

A. Multi-Step Onboarding Flow

- **Step 1: Personal Info** – Captures verified applicant bio-data, contacts, and passport photo.
- **Step 2: Academic Details** – Records secondary school data, graduation year, UTME scores, and 4 UTME subjects.
- **Step 3: Guardian Details** – Captures sponsor data and next-of-kin contacts.
- **Step 4: Final Review** – Summarizes all submitted data with instant score simulation prior to submission.

B. "Scan & Auto-fill" AI Document Scanner

Applicants upload their JAMB UTME Result Slip or O'Level Statement (PDF or image). With one click, Gemini Vision reads the file and automatically populates:

- Candidate Full Name & JAMB Registration Number
- Composite JAMB Score (100–400)
- Examination Year & Secondary School Name
- 4 UTME Subjects (normalized from acronyms like "PHY" to "Physics", "ENG" to "Use of English")

C. Deterministic 60/30/10 Admission Scoring Model

A transparent mathematical evaluation formula calculates candidate aggregate score out of 100%:

- **JAMB Component (60%):** $(\text{JAMB Score} / 400) * 60$
- **WAEC/NECO Component (30%):** Top 5 core subjects graded on a point scale (A1 = 6 pts down to C6 = 1 pt). $(\text{Total Grade Points} / 30) * 30$
- **Merit & Alignment (10%):** Single Sitting Bonus (+4 pts), Cutoff Surplus (+3 pts), and Core Subject Match (+3 pts).

- **Evaluation Categories:** $\geq 75\%$ = High Fit (Direct Merit Admit), 55–74% = Moderate Fit (Review Pool), $< 55\%$ = Low Fit.

D. Institutional Admin Decision Hub & Course Management

- **Admissions Decision Hub:** Allows admissions officers to view candidate dossiers, review mathematical breakdowns, inspect AI-generated recommendations, and execute verdicts (**Admit**, **Review**, or **Decline**).
 - **Course & Cutoff Control:** Enables department heads to add academic courses, adjust minimum UTME cutoffs, set student quotas, and monitor fill rates.
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5. Testing & Version Control

- **Linting & Code Quality:** ESLint with zero-tolerance policy (0 errors, 0 warnings across 270+ modules).
- **Version Control:** Git & GitHub branch workflows (feat/*, fix/*), Conventional Commits, and complete separation of API secrets via .gitignore and .env.example.
- **Build Performance:** Production build compiled via Vite in under 15 seconds, producing an optimized, compressed bundle (~378 kB gzip).
- **Deployment Readiness:** Pre-configured for single-command continuous deployment to Vercel

Testing

To test and inspect the Decision Hub, Applicant Dossiers, and Course Management features, use the configured credentials below:

Login Steps:

1. Open the login portal (/auth or click ****Sign In**** on the navigation bar).
2. Click the ****Admin / Partner**** tab.
3. Select the corresponding ****Institution**** from the dropdown menu.
4. Enter the email and password, then click ****Access Admin Dashboard****.

Admin Credentials

1. Platform Superadmin (Global System Access)

- Role: Platform Superadmin (Headquarters Control)
- Institution: ****ApplyNow Platform Administration (HQ)****
- Email: admin@applynow.ng

- Password: AdminSecret2026!
- Access Scope: Global oversight across all partner institutions, master candidate dossiers, and cross-university course controls.

2. Institutional Admin 1 (University of Lagos)

- Role: University Admissions Officer
- Institution: University of Lagos (UNILAG)
- Email: admin.unilag@applynow.ng
- Password: AdminSecret2026!
- Access Scope: UNILAG departmental applicants, course cutoffs, and admission decisions.

3. Institutional Admin 2 (FUTA)

- Role: University Admissions Officer
- Institution: Federal University of Technology, Akure (FUTA)
- Email: admin.futa@applynow.ng
- Password: AdminSecret2026!
- Access Scope: FUTA departmental applicants, course cutoffs, and admission decisions.

4. Institutional Admin 3 (University of Ibadan)

- Role: University Admissions Officer
- Institution: University of Ibadan (UI)
- Email: admin.ui@applynow.ng
- Password: AdminSecret2026!
- Access Scope: UI departmental applicants, course cutoffs, and admission decisions.

6. Conclusion & Future Enhancements

Conclusion

ApplyNow replaces outdated paper-based Post-UTME screening with an automated, AI-assisted platform. By combining instant OCR credential scanning, a fair and explainable scoring formula,

and a dedicated admin decision dashboard, the project dramatically reduces admission processing time from weeks to minutes.

Future Enhancements

1. **SMS & WhatsApp Alerts:** Automatic notification updates to candidates as their admission status changes.
2. **JAMB CAPS Sync:** Direct API integration with the national Central Admissions Processing System.
3. **Cloud Database Expansion:** Transition from client storage to distributed cloud databases (PostgreSQL / Cloud Firestore) with Role-Based Access Control.